

Shopify Integration

FastAPI · Next.js + React · MCP Server

You are being asked to build a Shopify integration using FastAPI (backend) and Next.js + React (frontend). This project connects to Shopify in two ways — through the Admin API to manage store data, and through the Storefront API to power a customer-facing shopping experience. There is also an MCP (Model Context Protocol) layer that exposes Shopify actions as AI-accessible tools.

T-01 Shopify API Configuration

Before anything else works, Shopify needs to know who is talking to it. Your job here is to make sure the backend is properly configured to communicate with Shopify. Once this is in place, all the other tasks will be able to use this connection to get things done.

T-02 Admin API

This is the management side of Shopify. You need to build a set of endpoints in FastAPI that let you perform actions on the Shopify store — adding a new product, updating an existing one, deleting a product, viewing orders, and managing customer records. Think of this as the backend control panel for the store.

T-03 Storefront

This is the customer-facing side of Shopify. You need to set up a way to fetch and display products from the Shopify store. Customers should be able to browse products, view product details, add items to a cart, and proceed to checkout. The data for all of this comes from Shopify.

T-04 MCP Server

Set up an MCP server that sits on top of the work done in T-02. It should expose the key Shopify actions — like searching for a product, fetching an order, or looking up a customer — so they can be accessed as tools. The MCP server should not repeat any logic already written; it simply connects to what is already there.

T-05

Frontend UI

Build the pages that customers will actually see and use. This includes a page that lists all products, a page for each individual product, a cart that updates as customers add or remove items, and a checkout page. Everything on these pages should reflect real data coming from Shopify.

Complete all five tasks and submit your code with a short README on how to run it locally.